Diamond Lemma and its Applications

MTH391 Project Report

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1 Abstract

The Diamond Lemma is a combinatorial result that provides a criterion for deciding the unambiguity in the process of applying reductions on a set by means of pre-defined binary relations on the set. Although the final result of the lemma is quite intuitive and may seem far from interesting at a first glance, setting it down formally with a rigorous proof requires some amount of effort and gives us an insight into how helpful a tool it can be in solving problems in various realms of mathematics.

In this report, we'll first have a look at a problem that will provide us some motivation for this theorem. Next, we'll be looking at 2 versions of Diamond Lemma, one by Newman and the other by Bergman. Finally, we'll see some applications of the lemma to problems in Combinatorics, Lie Algebras, Matrix Theory and Gröbner Base Theory for commutative polynomial rings.

Most of the pre-requisites have been discussed and can also be read up from the references. It is assumed that the reader has a good command over rigorous proofs and arguments and a working knowledge of terminology in algebra.

2 A motivational problem

We'll be discussing a problem of combinatorial nature that will outline the 'diamond' condition and will serve as a proof strategy for a majority of our theorems and problems. The source of this problem is ⟨https://sbseminar.wordpress.com/2009/11/20/the-diamond-lemma/⟩.

Problem 2.1. *Suppose we have a finite collection of great circles on a sphere, none of them through the north pole or the south pole. Let R be the set of regions in the complement of these circles, and suppose that every region is a triangle. Put a partial order on R by $x \leq y$ if x is south of every circle that y is south of. Show that, for x and $y \in R$, there is some $z \in R$ such that $w \leq z$ if and only if $w \leq x$ and $w \leq y$.*

For any x, y in R, if such a z exists, we refer to it as the GLB (greatest lower bound) of x and y. (easy to see that such a z has to be unique) We'll try to attack this problem using induction, trying intuitively to go from bottom to top and finding GLB's for each pair. In order to formalise this, for any region r, we let $l(r)$ be the number of circles below r. We will prove the following claim by induction on $l(r)$ - for any $x \leq r$ and $y \leq r$, GLB of x and y exists.

The claim is trivially true for $l(r) = 0$ establishing the base case. Now, consider r and some $x \leq r$ and $y \leq r$. Consider southward going paths $r \to x_1 \to \ldots \to x$ and $r \to y_1 \to \ldots \to y$. If $x_1 = y_1$, then we can rename it as r and use the induction assumption to reach the required conclusion. If not, these must exist a region adjacent to both of them and towards their south, say z. One can verify that z is the GLB of x_1 and y_1.

Now, consider $z \leq x_1$ and $x \leq x_1$. As $x_1 < r$, by induction assumption, z and x have an GLB, say z_1. Similarly, z and y also have an GLB, say z_2. Finally, $z_1 \leq z$ and $z_2 \leq z$, and so, as $z < r$, again by induction assumption, we have a GLB w of z_1 and z_2. Now, it is a matter of verifying our definitions to see that w is the GLB of x and y. Thus, by induction, our claim stands proven,

Considering the region r containing the north pole, and applying our claim on it using any 2 regions x and y, we see that x and y must have an GLB and so, we have solved our problem.

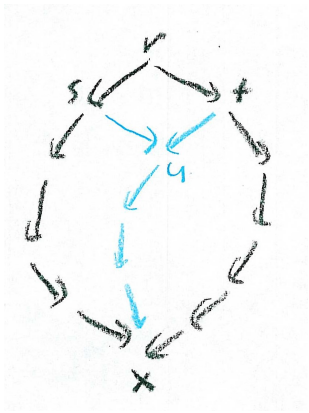


Figure 1: Diamond condition

The above figure very well conveys the idea of the proof strategy we just used for this problem and which will be used time and again in this report. This is the diamond structure from which the diamond lemma derives its name and situations which can be geometrically depicted as above are said to satisfy the diamond condition.

3 Diamond Lemmas

We are going to deal with 2 versions of the diamond lemma. The first one is by Newman and was given by him initially in the form of graph theoretic terminology. The second one is by Bergman which uses ring theoretic terminology. It will be seen that the 2nd one is a strengthening of the first one as it provides relaxation of some conditions. After this section, whenever we talk about the diamond lemma, we'll exclusively refer to Bergman's version.

The content for Newman's version is from [4] while for Bergman's version is from [2].

3.1 Newman's Diamond Lemma

Given a non-empty set S, we define a reduction system on S to be a relation R on S such that R is strictly antisymmetric, i.e., for any $a, b \in S$, $(a, b) \in R \Rightarrow (b, a) \notin R$.

We use the notation $a \to b \Leftrightarrow (a, b) \in R$. If this happens, we say that there is an edge from a to b. Thus, defining edges like this on the vertex set S, we have graph. We say $a \to^n b$ if there is a path from a to b of length n. We say $a \to^* b$ if there is a path from a to b.

Next we use the notation $a \leftrightarrow b \Leftrightarrow (a, b) \in R$ or $(b, a) \in R$. As above, we can define $a \leftrightarrow^n b$ iff there exist $n - 1$ vertices $v_1, v_2, \ldots v_{n-1}$ such that $a \leftrightarrow v_1 \leftrightarrow v_2 \ldots \leftrightarrow v_{n-1} \leftrightarrow b$. Similarly, we define $a \leftrightarrow^* b$ iff $a \leftrightarrow^n b$ for some n. It is easy to see that $\leftrightarrow^*$ defines an equivalence relation on S.

We say that $a \in S$ is in normal form if there are no outgoing edges from a. We say that b is a normal form of a if b is in normal form and there exists a path from a to b. Next, we say that $\to$ satisfies DCC (descending chain condition) if there exists no infinite chain of reductions. If this is the case, it is clear from the definition that for every a, there exist a normal form of a in S. With this we are ready to tackle the results due to Newman

Theorem 3.1. *Let a reduction system $\to$ be defined on a set S such that it*

satisfies DCC. Then, the following are equivalent:

1. *Local confluence (Diamond condition) : If $a \to b$ and $a \to c$, then there exists d such that b and c can be reduced to d.*
2. *Confluence : If $a \to^* b$ and $a \to^* c$, then there exists d such that b and c can be reduced to d.*
3. *Every element in S has a unique normal form.*
4. *Church-Rosser property If $a \leftrightarrow^* b$, then then there exists d such that a and b can be reduced to d.*

Proof. Indexing the above statements, we'll show that (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2) : Suppose $a \to^* b$ and $a \to^* c$, i.e. there are paths from a to b and c. Then the claim (2) follows by induction on the sum of the lengths of these 2 paths where (1) serves as the base case and in the general case, helps us in reducing the size of our problem to make the induction assumption applicable. Here, we make use of the fact that every element can be reduced to a normal form.

(2) $\to$ (3) : Suppose there is an element a which can be reduced to 2 normal forms b and c. Then, we have that $a \to^* b$ and $a \to^* c$ and so, by claim (2), b and c must both be reducible to some d. But, as both of them are in normal form, we get that $b = c = d$, thus proving (3).

(3) $\to$ (4) : Suppose $a \leftrightarrow^* b$. Hence, there exist n vertices $v_1, v_2, \ldots v_{n-1}$ such that $a \leftrightarrow v_1 \leftrightarrow v_2 \ldots \leftrightarrow v_n \leftrightarrow b$. $a \leftrightarrow v_1$ means that one of them can be reduced to the other. By claim (3), we get that a and v_1 can be reduced to the same normal form. Similarly, v_1 and v_2 can be reduced to the same normal form, and so on. Thus, we get that a and b can be reduced to the same normal form thereby proving (4).

(4) $\to$ (1) : If $a \to b$ and $a \to c$, then we have that $b \leftrightarrow^* c$ and so, by (4), b and c can be reduced to the same normal form.

□

Corollary 3.2. *(Newman's Diamond Lemma) Let a reduction system $\rightarrow$ be defined on a set S such that it satisfies DCC and the diamond condition. Then, every equivalence class of $\leftrightarrow^*$ consists of a unique normal form.*

3.2 Bergman's Diamond Lemma

Consider a commutative associative ring k with unity. Consider a set X and let the semigroup generated by it be $\langle X \rangle$. Let $k\langle X \rangle$ denote the corresponding free associative k-algebra. Next consider a set S that consists of ordered pairs of the form $\sigma = (W_\sigma, f_\sigma)$ where $W_\sigma \in \langle X \rangle$ and $f_\sigma \in k\langle X \rangle$.

Next, for each $\sigma \in S$ and $A, B \in \langle X \rangle$, define a k-module map $r_{A\sigma B}$ that sends $AW_\sigma B$ to $Af_\sigma B$ and fixes all other monomials. Each of these maps is known as reduction. An elements $a \in k\langle X \rangle$ is said to be irreducible if all the reductions act trivially on it. All the irreducible elements form a submodule of $k\langle X \rangle$ denoted by $k\langle X \rangle_{irr}$.

An element a of $k\langle X \rangle$ is said to be reduction finite if for every infinite sequence of reductions $r_1, r_2, \ldots$, there exists an n such that for all $m > n$, r_m acts trivially on $r_{m-1}(\ldots r_2(r_1(a))$. Reduction finite elements too form a submodule of $k\langle X \rangle$. a is said to be reduction unique if it is reduction finite and if it can be reduced to a unique irreducible element. This irreducible element is denoted by $r_s(a)$. We have the following lemma that describes some properties of reduction unique elements.

Lemma 3.3. *1. The set of reduction unique elements forms a submodule of $k\langle X \rangle$, and r_s is a k-linear map from this module onto $k\langle X \rangle_{irr}$.*

2. Given $a, b, c \in k\langle X \rangle$, such that their product abc is a k-linear sum of reduction unique monomials. Let r be any reduction. Then $r_s(ar(b)c) = r_s(abc)$.

The proof of this lemma is just a matter of routine verifications and has been omitted.

Next, we consider a 5-tuple (σ, τ, A, B, C) with $\sigma, \tau \in S$ and $A, B, C \in \langle X \rangle$ such that $W_\sigma = AB$ and $W_\tau = BC$. Then, we call (σ, τ, A, B, C) an overlap ambiguity. This ambiguity is said to be resolvable if there exist compositions of reductions r and r' such that $r(f_\sigma C) = r'(Af_\sigma)$. (This is known as the confluence condition).

Similarly, a tuple (σ, τ, A, B, C) with $\sigma, \tau \in S$ and $A, B, C \in \langle X \rangle$ such that $W_\sigma = ABC$ and $W_\tau = B$ is said to be an inclusion ambiguity. Its resolvability can be defined in a similar manner.

Now, let '$\leq$' be a semigroup partial order on $\langle X \rangle$ i.e. for any $A, B, C, D \in \langle X \rangle$, $B \leq D \Rightarrow ABC \leq ADC$. This order is said to be compatible with S if for all $\sigma \in S$, f_σ is k-linear combination of monomials $< W_\sigma$. Next, define I to be the 2-sided ideal generated by elements of the form $W_\sigma - f_\sigma$.

Now, we define something to act as a local confluence condition. For any $A \in \langle X \rangle$, define I_A to be a submodule of $k\langle X \rangle$ spanned by elements of the form $B(W_\sigma - f_\sigma)C$ where $W_\sigma < A$. We say that an overlap ambiguity (σ, τ, A, B, C) is resolvable relative to '$\leq$' iff $f_\sigma C - Af_\tau \in I_{ABC}$. (A similar definition can be given for inclusion ambiguities).

With this, we are ready to attack the main theorem of this report.

Theorem 3.4. *(Diamond Lemma) Let S be a reduction system for a free associative algebra $k\langle X \rangle$ and $\leq$ be a semigroup partial ordering that is compatible with S and has descending chain condition. Then, the following are equivalent:*

1. *All ambiguities of S are resolvable*
2. *All ambiguities of S are resolvable relative to $\leq$*
3. *All elements of $k\langle X \rangle$ are reduction unique under this reduction system.*
4. *A set of representatives for the algebra $R = k\langle X \rangle / I$ is given by the k-submodule $k\langle X \rangle_{irr}$ or equivalently, $k\langle X \rangle = k\langle X \rangle_{irr} \oplus I$.*

Proof. We'll establish the equivalence by showing that (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (3) $\Rightarrow$ (1).

(3) $\Rightarrow$ (4) : If all elements are reduction unique, r_s becomes a map from $k\langle X\rangle$ to $k\langle X\rangle_{irr}$ whose kernel is given by I. Therefore, we get that $k\langle X\rangle/I \cong k\langle X\rangle_{irr}$ which implies (4).

(4) $\Rightarrow$ (3) : Suppose an element a can be reduced to 2 irreducible values b, $b' \in k\langle X\rangle_{irr}$. Then it can be seen that $b - b' \in k\langle X\rangle_{irr} \cap I = \{0\}$. Thus, $b = b'$ and a is reduction unique.

(3) $\Rightarrow$ (1) : Obvious

(1) $\Rightarrow$ (2) : Consider a resolvable overlap ambiguity (σ, τ, A, B, C). (inclusion ambiguities can be analogously handled) Then, we know that $W_\sigma C$ and AW_τ are reducible to a common element R. Then, it is easy to see that $R - W_\sigma C \in I_{ABC}$ and $R - AW_\tau \in I_{ABC}$. Subtracting these 2, we get the desired result.

(2) $\Rightarrow$ (3) : It is sufficient to show that all the monomials are reduction unique because k-linear combinations of monomials generates $k\langle X\rangle$. Now, as $\leq$ satisfies descending chain condition, every element has to be reduction finite. Suppose we are given a monomial D. Inductively, let us assume that all monomials $< D$ are reduction unique and try to establish the reduction uniqueness of D. Suppose there are 2 possible reductions of D say $r_{L\sigma M'}$ and $r_{L'\sigma M}$. Then, we have the following cases: (without loss of generality, suppose L' is longer than L)

1. The sub-words W_σ and W_τ overlap but neither contains the other. This is an overlap ambiguity. That is, D is of the form $LABCM$ and the ambiguity is (σ, τ, A, B, C). Now, by (2), $f_\sigma C - Af_\tau \in I_{ABC}$ and so, $L(f_\sigma C - Af_\tau)M \in I_{LABCM} = I_D$. Now, as all monomials less than D are reduction unique, we must have that I_D vanishes under the map r_s. Thus, $r_s(L(f_\sigma C - Af_\tau)M) = 0$ and by linearity, we get that both the reduced forms of D have the same value of r_s. Thus, D is reduction

unique.

2. One of the subwords contains the other. This is the case of an inclusion ambiguity and can be handled in exactly the same manner as the previous case.

3. The subwords are disjoint. In this case, we can apply $r_{L\sigma M'}$ on $r_{L'\sigma M}(D)$ and $r_{L'\sigma M}$ on $r_{L\sigma M'}(D)$ to get the same expression and so, by lemma 3.3, we get that D is reduction unique.

□

The fact that conditions (1) and (2) of the theorem are equivalent shows that if we can verify either of these for a given set of reductions, we can apply this theorem to get reduction-uniqueness as a result. The theorem requires that we need to only check that the minimal nontrivial ambiguously resolvable monomials are reduction unique to get reduction uniqueness for the entire algebra.

4 Applications

4.1 Poincare Birkhoff Witt Theorem

Let k be a commutative ring and consider a Lie algebra L over k that is free as a k-module, with basis X. We can consider the associative free algebra $k\langle X\rangle$ of which L is a submodule. Now, consider the 2-sided ideal I generated by elements of the form $ab - ba - [a, b]$ where $a, b \in L$.

Now, consider the ring $k[L] = k\langle X\rangle/I$. This is known as the universal enveloping algebra of the Lie algebra L. It is so named because of the following reason. Let ϕ denote the canonical Lie algebra homomorphism from L to $k[L]$. Suppose we are given an associative k-algebra R such that we have k-linear map ψ from L to R such that $\psi([a, b]) = \psi(a)\psi(b) - \psi(b)\psi(a)$. Then, there exists a unique algebra homomorphism ρ from $k[L]$ to R such that $\psi = \rho \circ \phi$.

However, this definition provides us very little insight about the structure of $k[L]$. For this purpose, we have the following theorem which provides a description of a basis of $k[L]$ when looked upon as a k-module. The proof that we mention here has been taken from [2].

Before that, let $\leq$ be a total order on X. Then, we can assume that I is generated by elements of the form $ab - ba - [a, b]$ such that $a < b$.

Theorem 4.1. *(Poincare Birkhoff Witt) $k[L]$ is free as a k-module with basis consisting of elements of the form $x_1x_2 \ . \ . \ . \ x_n$ such that $x_i \in X$ for all i and $x_1 \leq x_2 \leq \ . \ . \ . \leq x_n$*

Proof. $k[L]$ is generated as a k-module by elements of the form $R + I$ where R is a monomial consisting of variables from X. We'll show that these are generated by elements of the form as mentioned in the statement of the theorem. Consider the set S of reductions consisting of the ordered pairs - $(yx, xy - [x, y])$ whenever $y > x$. (which makes sense as $xy - yx - [x, y] = 0$ in $k[L]$ as these elements have been quotient-ed out).

Now, it is easy to see that the irreducible elements under this set of reductions is exactly the set of elements which we wish to prove form the basis. Thus, it is sufficient to prove that every reducible element can be reduced under the set of reductions to a unique irreducible element generated by this basis. For that, we need to check 2 things - descending chain condition and resolution of ambiguities.

In order to establish DCC, we define $inv(x_1, x_2, \ . \ . \ . \ x_n)$ for any $x_1x_2 \ . \ . \ . \ x_n \in \langle X \rangle$ to be the numbered of order pairs (i, j) such that $i > j$ but $x_i < x_j$. Using this, we can define a partial order on $\langle X \rangle$ as follows: set $A < B$ if $length(A) < length(B)$ or if $length(A) = length(B)$ and $inv(A) < inv(B)$. Then, it can be seen that $\leq$ is a semigroup partial ordering compatible with S and having descending chain condition as the values of $length$ and inv can at most go down to 1 and 0 respectively.

Next, the only ambiguities that we need to check for are of the form $(\sigma_{zy}, \sigma_{yx}, z, y, x)$ where $z > y > x$.

$$r_{1\sigma_{zy}x}(zyx) - r_{z\sigma_{yx}1}(zyx) = (yzx - [y, z]x) - (zxy - z[x, y])$$

To get to an irreducible form, we need to apply $r_{y\sigma_{zx}1}$ and $r_{1\sigma_{yx}z}$ in that order on yzx and $r_{1\sigma_{zx}y}$ and $r_{x\sigma_{zy}1}$ in that order on zxy. The result after the application of these 4 reductions is:

$$(x[y, z] - [y, z]x) + (y[z, x] - [z, x]y) + (z[x, y] - [x, y]z)$$

Now, $x[y, z]-[y, z]x-[x, [y, z]] \in I_{zyx}$ as $length(x[y, z]) = length([y, z]x) = 2 < 3 = length(zyx)$. Writing similar expressions for the other terms, we get that module I_{zyx}, the above expression is congruent to:

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]]$$

which is equal to 0 by Jacobi identity. Thus, we get that $r_{1\sigma_{zy}x}(zyx) - r_{z\sigma_{yx}1}(zyx) \in I_{zyx}$ and so, by definition, the ambiguity is resolvable relative to $\leq$. Thus, the conditions of diamond lemma have been met and so, the theorem stands proven.

□

4.2 A combinatorial problem

Consider the 2-dimensional $X-Y$ plane in Cartesian coordinates with all its lattice points. Draw all the lines passing through the lattice points that are parallel to either of the coordinate axes. We'll refer to these as lattice lines. Suppose h and k are 2 non-negative integers. Then, what are the number of ways of going from $(0,0)$ to (h,k) such that we move only along lattice lines and our each move occurs in either the positive X-direction (which we'll denote by R) or the positive Y-direction (which we'll denote by U)?

So, this is a quite standard problem in elementary counting theory and the answer is $\binom{h+k}{k}$. We'll try to visualise this solution with the help of the binomial theorem. Consider:

$$(R+U)^{h+k} = (R+U)(R+U)\ .\ .\ .\ (R+U)$$

Any move sequence consisting of $h+k$ moves that consists of exactly h R's and k U's, which denote unit movements in the right and the up directions respectively, will take us from $(0,0)$ to (h,k). Any such sequence can be visualised as being obtained by choosing either R's or U's from each term of the RHS of the above equation such that we ensure that we pick the required number of each. Hence, the number of such sequences will equal the coefficient of R^hU^k in the above expansion, which is equal to $\binom{h+k}{k}$ by the binomial theorem.

Now, let us look at a variant of this problem. Suppose we still want to go from $(0,0)$ to (h,k) using the same rules, but with the added rule that the area enclosed between our path and the X-axis should be exactly j units where j is an integer satisfying $0 \leq j \leq hk$. How many such paths are there now? If we take $j=0$, then there's only one such path - go from $(0,0)$ to $(h,0)$ and then to (h,k). How do we solve the problem for a general j? Let us look at:

$$(R+U)^{h+k} = (R+U)(R+U)\ .\ .\ .\ (R+U)$$

In the earlier case of the problem, it didn't matter if we went up first and then right or vice-versa, the order didn't affect our calculations. In algebraic terms, earlier we had $RU = UR$ and so we had been able to use our

traditional binomial theorem. In this problem, RU and UR will be different because these paths enclose different areas beneath them. Let us consider the reductions - $UR = qRU$, $Rq = qR$ and $Uq = qU$ where q is a dummy variable (to account for the non-commutativity). Using these, given any string consisting of U's and R's, it can be reduced to a canonical form $q^t.R^uU^v$ where $t, u, v \in \mathbb{Z}^+ \cup \{0\}$. For example, $URURUU = URq.RUUU = Uq.RRUUU = q.URRUUU = q.q.RURUUU = q^2.RURUUU = q^2.Rq.RUUUU = q^2.q.RRUUUU = q^3.R^2U^4$.

The above string corresponds to a move from $(0, 0)$ to $(2, 4)$ where the area enclosed between the path and the X-axis is 3 units. Looking at the reduction process geometrically, every time we replace UR by RU, the area enclosed gets reduced by 1 and at the same time, a q gets introduced in the term. Therefore, it is easy to see that if every string of U's and R's can be reduced to a unique such canonical form, our problem will be solved.

In order to show this, we check that all the conditions of diamond lemma are applicable. Defining a total order on the 3 symbols as : $q < R < U$. Then, we can define a partial monomial order in exactly the same fashion as the previous problem to get that the partial order satisfies descending chain condition. Next, we need to ensure that the ambiguities are resolvable. The only ambiguous case is the resolution of URq which can be reduced to either $qRUq$ or to UqR. In the first case, we have the chain of reductions $qRUq \rightarrow qRqU \rightarrow qqRU = q^2RU$. In the 2nd case, we have $UqR \rightarrow qUR \rightarrow qqRU = q^2RU$. Therefore, we see that the ambiguity is resolvable. Hence, we can apply the diamond lemma to get that every string is reducible to a unique canonical form.

Hence, we can can conclude that if we expand $(R + U)^n$ using the above reduction rules that we defined, then the coefficient of $q^aR^bU^c$, where $b+c = n$, denotes the number of paths from $(0, 0)$ to (b, c) that enclose an area of a units below the path. In order to find this coefficient, we apply the q-binomial theorem: (see [3])

$$(R+U)^n = \sum_{k=0}^{n} \binom{n}{k}_q R^{n-k}U^k$$

where:
$\binom{n}{k}_q = \frac{n!_q}{k!_q(n-k)!_q}$

$n!_q = [n]_q[n-1]_q \ . \ . \ . \ [2]_q[1]_q$

$[n]_q = 1 + q + q^2 + \ . \ . \ . \ + q^{n-1}$

In order to demonstrate the working of the solution, I state an example. Suppose we wish to go from $(0,0)$ to $(4,5)$ with an area of 12 units enclosed by the path. By the q-binomial theorem, we have:

$$(R+U)^9 = \sum_{k=0}^{9} \binom{9}{k}_q R^{n-k}U^k$$

Coefficient of R^4U^5 is given by $\binom{9}{4}_q = (1-q^9)(1-q^8)(1-q^7)(1-q^6)/(1-q^4)(1-q^3)(1-q^2)(1-q) = q^{20}+q^{19}+2q^{18}+3q^{17}+5q^{16}+6q^{15}+8q^{14}+9q^{13}+11q^{12}+11q^{11}+12q^{10}+11q^9+11q^8+9q^7+8q^6+6q^5+5q^4+3q^3+2q^2+q+1$. Thus, the required answer is the coefficient of $q^{12}R^4U^5 = 11$. Therefore, the number of paths is 11.

4.3 Amitsur-Levitski Theorem

The following is a combinatorial theorem in matrix theory which comes as quite a surprise. The proof that we are going to look at is from [1]. Before moving towards the theorem, we introduce some notation. For variables $x_1, x_2, \ldots, x_n$, we define:

$$St_n(x_1, x_2, ..., x_n) = \sum_{\sigma \in S_n} sign(\sigma) x_{\sigma(1)} x_{\sigma(2)} ... x_{\sigma(n)}$$

where S_n is the symmetric group of order n. In the proof, we are going to use the following elementary results whose proofs have been omitted.

Lemma 4.2. *Let R be a commutative algebra over $\mathbb{Q}$ and $A \in M_n(R)$. Suppose $tr(A^i) = 0$ for all $i > 0$. Then $A^n = 0$.*

Lemma 4.3. *If r is an even natural number, then*

$$tr(St_r(A_1, A_2, ..., A_r)) = 0$$

for all $A_1, A_2, \ldots, A_r \in M_n(R)$.

Lemma 4.2 can be proved by realising that the trace of a mtrix is the sum of its eigenvalues whereas, Lemma 4.3 follows from the fact that for 2 square matrics A and B, we have $tr(AB) = tr(BA)$.

Theorem 4.4. *(Amitsur-Levitski) Let R be a commutative ring, $n > 0$ be a natural number and $M_n(R)$ denote the ring of $n \times n$ matrices over R. Then, for all $A_1, A_2, \ldots, A_{2n} \in M_n(R)$:*

$$St_{2n}(A_1, A_2, ..., A_{2n}) = 0$$

Proof. Consider $2n^2$ transcendentals x_{ijk}, $1 \leq i, j, \leq n, 1 \leq k \leq 2n$ and let K be the polynomial ring of these variables over $\mathbb{Q}$. Next, consider matrices $B_1, B_2, \ldots, B_{2n} \in M_n(K)$ such that $B_k = [x_{ijk}]_{i,j=1}^n$. Now, consider the free algebra $F = K < Y >$ where $Y = \{y_1, y_2, \ldots, y_{2n}\}$. Next, consider the 2 sided ideal I of F generated by elements of the form $y_i y_j + y_j y_i$ and consider the quotient algebra $D = F/I$. Introduce the notation $e_i := y_i + I \in D$.

Now, we'll use the diamond lemma to find a basis for D when looked upon as

a K-module. Any element of D is of the form $f + I$ where f is a polynomial in the y_i's over K. In order to have a canonical form for each of these elements, we introduce a system of reductions of the form $y_j y_i = -y_i y_j$ where $i \leq j$. (which makes sense because elements of the form $y_i y_j + y_j y_i$ have been quotient-ed out and are equal to 0 in D) Again using a similar partial order on the monomials that we have been using till now, by defining $y_1 \leq y_2 \leq \ . \ . \ . \ \leq y_{2n}$, it can be seen that DCC is satisfied and the ambiguities are resolvable and so, by diamond lemma, each monomial consisting of the y_i's can be reduced to the canonical form $e_{i_1} e_{i_2} . \ . \ . e_{i_k}$ where $k \leq 2n$, such that $i_1 < i_2 < \ . \ . \ . \ i_k$. These elements along with 1 form a K-module basis for D.

Now, let $B = B_1 e_1 + B_2 e_2 + \ . \ . \ . \ + B_{2n} e_{2n} \in M_n(D)$. Then, a simple computation gives that:

$$B^k = \sum_{i_1 < i_2 < \ldots < i_k} St_k(B_{i_1}, B_{i_2}, ..., B_{i_k}) e_{i_1} e_{i_2} ... e_{i_k}$$

for $k \leq 2n$ and $B^k = 0$ otherwise. For any $r > 0$, using the above expansion for B^{2r} and Lemma 4.3, we get that $tr(B^{2r}) = 0$. As this is true for all r, Lemma 4.2 gives us that $B^{2n} = 0$. By the above expansion, $B^{2n} = St_{2n}(B_1, B_2, \ . \ . \ . \ B_{2n}) e_1 e_2 \ . \ . \ . \ e_{2n}$. Thus, we must have $St_{2n}(B_1, B_2, \ . \ . \ . \ B_{2n}) = 0$.

Now, let L be the polynomial ring of the variables x_{ijk}'s over $\mathbb{Z}$. Then, each $B_i \in M_n(L)$. Now, consider a map ϕ from L to R such that $\phi(1_L) = 1_R$ and $\phi(x_{ijk})$ is assigned such that if ϕ' is the extension of this map to a map from $M_n(L)$ to $M_n(R)$, for each i, we have $\phi'(B_i) = A_i$. Then, as $\phi'(0) = 0$, we have $St_{2n}(A_1, A_2, \ . \ . \ . \ A_{2n}) = 0$ completing the proof of the theorem. $\square$

4.4 Characterisation of Gröbner Bases

In this section, we'll be looking at a specific category of bases of commutative polynomial rings called Gröbner bases and use the diamond lemma to find a necessary and sufficient criterion for a basis to be a Gröbner basis. The content of this section is referred to from [4].

Let k be a field and $k[x_1, x_2, \ldots, x_n]$ denote the commutative polynomial ring of n variables over k. Let M denote the set of all monomials. Every $f \in k[x_1, x_2, \ldots, x_n]$ can be expressed uniquely (upto a permutation of the monomials) in the form $f = \sum_{i=1}^{t} c_i m_i$ where, for all i, $c_i \in k$ and $m_i \in M$.

Suppose $\leq$ is a total order on M. Then, the above expression for f can me made such that we have $m_1 > m_2 > \ldots > m_t$. $c_1 m_1$ is known as the leading term $lt(f)$, m_1 is known as the leading monomial $lm(f)$ and c_1 is known as the leading coefficient $lc(f)$.

Now, given any subset G of $k[x_1, x_2, \ldots, x_n]$, define the leading monomial ideal of G, $lm(G)$ to be the 2-sided ideal generated by elements of the form $lm(g)$ for $g \in G$. Given an ideal I, if we have a finite subset G of I, such that $lm(G) = lm(I)$, then G is referred to as a Gröbner basis of I.

With just this definition at hand, we can't say anything about the existence of Gröbner bases or their properties. The aim of the rest of the section is to throw some light over these issues. Our first aim is to prove the existence of a Gröbner basis for any ideal I. This will be established through the help of Hilbert Basis Theorem and Dickson's Lemma.

A ring R is said to be Noetherian if it satisfies the ascending chain condition i.e. there is no infinite properly ascending chain of ideals. Then, it is a simple exercise to establish that a ring R is Noetherian if and only if every ideal of R is finitely generated. Next, we state the Hilbert Basis Theorem which is a standard result and so, its proof has been omitted.

Theorem 4.5. *(Hilbert Basis Theorem) If a ring R is Noetherian, so is the ring $R[x]$.*

An easy corollary of this theorem is that whenever R is Noetherian, $R[x_1, x_2, \ldots, x_n]$ is also Noetherian. Now, as k is a field, it is trivially Noetherian and so, $k[x_1, x_2, \ldots, x_n]$ is also Noetherian and thus, all of its ideals are finitely generated. In fact, given any ideal I generated by a subset S, we can find a finite subset S' of S such that $I =< S >=< S' >$. Thus, we have the following important result.

Theorem 4.6. *(Dickson's Lemma) Let S be a non-empty set of monomials of $k[x_1, x_2, \ldots, x_n]$. Then there exists a finite subset S' of S such that $< S >=< S' >$. Equivalently, there exists a finite S' which is a subset of S such that for every $m \in S$, there exists $m' \in S'$ such that m' divides m.*

This lemma establishes the existence of Gröbner bases because given any ideal I, we can find a finite subset S of I such that $lm(S) = lm(I)$. The following result provides an interesting property of Gröbner bases.

Theorem 4.7. *Suppose I is an ideal of $k[x_1, x_2, \ldots, x_n]$ such that G is a Gröbner basis. Then, $I =< G >$.*

Proof. Let $f \in I$. Put $f_1 = f$. Then, $lm(f_1) \in lm(I) = lm(G)$. Hence, there exists $g_1 \in G$, $m_1 \in M$ and $c_1 \in k$ such that $lt(f_1) = lt(c_1 g_1 m_1)$. Let $f_2 = f_1 - c_1 g_1 m_1 \in I$. Repeat this procedure over and over. As during each step, the leading monomial of the polynomial decreases, the process has to eventually terminate and so, there exists an n such that $f_n = 0$. Therefore,

$$0 = f_n = f_{n-1} - c_{n-1} g_{n-1} m_{n-1} = \ldots = f_1 - \sum_{i=1}^{n-1} c_i g_i m_i$$

And so, $f(= f_1)$ can be generated by the g_i's and thus $f \in< G >$. As this is true for all $f \in I$, we get that $I =< G >$. □

Now that we have proven their existence and have some of their properties, we wish to characterize the Gröbner bases. In order to do this, we'll be making use of the diamond lemma and so, we need to introduce some terminology to prepare a framework where the lemma can be applied.

Given 2 polynomials f and g, if $lm(g)$ divides some non-zero term cm in f, let $h = f - \frac{cm}{lt(g)} g$. This replaces the term cm in f by a k-linear combination of monomials $< m$. We refer to this process as a polynomial reduction

and say that f reduces to h modulo g. Similarly, given a finite set G of polynomials, we say that a polynomial f can be reduced to h modulo G if there exists a sequence of polynomial reductions modulo polynomials that occur in G that turns f to h.

If a polynomial r can not be reduced modulo G, we call it an irreducible polynomial with respect to the sequence of reductions. If for a polynomial f, we have unique reduced form module G, we denote it by $R(f, G)$. Summing it all up, given polynomials f and r such that r is a reduced form of f with respect to a set of polynomials G, then either $f = r$ or there exists s such that $f = \sum_{i=1}^{s} c_i m_i g_i + r$ where the c_i's are in k, m_i's are in M and g_i's are in G.

At this stage, it is not at all clear if a given set G can be used to associate a well-defined procedure for reducing a polynomial that provides us with a unique reduced form for each polynomial. The following theorem demonstrates a connection between the sets G for which this can be done and Gröbner bases and establishes conditions which ensure the uniqueness of the reduced form of polynomials. It also provides us with a criterion that helps us in checking if a given set is a Gröbner basis for an ideal I.

Theorem 4.8. *Let $G = \{g_1, g_2, \ldots, g_l\}$ be a finite subset of $k[x_1, x_2, \ldots, x_n]$. Let $I =< G >$ and let $\leq$ denote a monomial order on M. Then, the following are equivalent:*

1. *$lm(G) = lm(I)$*
2. *For all $f \in I$, $lm(g_i) | lm(f)$ for some i*
3. *$f \in I \Leftrightarrow R(f, G) = 0$*
4. *For all polynomials f, the reduced form of f with respect to G is unique.*
5. *As k-vector spaces, $k[x_1, x_2, \ldots, x_n] = k_R(G) \oplus I$*
6. *For all $g_1, g_2 \in G$, $R(S(g_1, g_2), G) = 0$*

Here, $k_R(G)$ denotes the set of all reduced polynomials and $S(g_1, g_2) = \frac{L}{lt(g_1)} g_1 - \frac{L}{lt(g_2)} g_2$ is known as the S-polynomial of g_1 and g_2, where L is the LCM of $lt(g_1)$ and $lt(g_2)$.

Proof. The theorem will be proven if it can be shown that (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (6) $\Rightarrow$ (4) and (5) $\Rightarrow$ (1). We'll be showing that (6) $\Rightarrow$ (4) as the rest are easily verified.

Consider any polynomial $f \in k[x_1, x_2, \ldots, x_n]$. Suppose there exist 2 polynomials g_1 and g_2 modulo which a single monomial M of f can be reduced. Then, M must be a multiple of L, the LCM of the leading terms of g_1 and g_2. On the application of these reductions, f turns to $f - \frac{M}{lt(g_1)}g_1$ and $f - \frac{M}{lt(g_2)}g_2$ respectively. These 2 reduced forms differ by $\frac{M}{lt(g_2)}g_2 - \frac{M}{lt(g_1)}g_1 = \frac{M}{L}S(g_1, g_2)$. Now, if we assume (6), $S(g_1, g_2)$, and as a consequence $\frac{M}{L}S(g_1, g_2)$ too, is reducible to 0 modulo G. Hence, by linearity, it can be seen that both of the above reduced forms of f can be reduced to the same irreducible form.

So what we have is this. We have a monomial partial ordering $\leq$ that is compatible with the set of reductions G and also, all ambiguities are resolvable (as shown in the previous paragraph). Therefore, by applying Diamond Lemma, we get that every element of the polynomial ring must be reducible to a unique irreducible element and hence, (4) is true. Therefore, (6) $\Rightarrow$ (4). $\square$

This theorem in essence proves that a set of reductions G has no ambiguities if and only if G forms a Gröbner basis of the polynomial ring. Further, both of these are equivalent to the fact that the S-polynomial for any ordered pair consisting of elements of G is reducible to 0 modulo G. This fact forms the crux of Buchberger's algorithm for computing Gröbner bases for commutative polynomial rings(see [4]).

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DECLARATION

I hereby declare that the work presented in the project report titled "Diamond Lemma and its Applications" contains my own ideas in my own words. At places, where ideas and words are borrowed from other sources, proper references, as applicable, have been cited. To the best of my knowledge this work does not emanate or resemble to other work created by anyone other than mentioned herein. The work was created on the 3rd of November, 2015.

Name and Signature:

Date: